

LAB 01

Digital RTL Design

Combinational Circuit Design

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1 MUX_2_1

```
1 module mux2 (in0,in1,sel,out);
2     input in0,in1,sel;
3     output out;
4
5     assign out = (sel==1)? in1 : in0;
6 endmodule
```

2 Priority Encoder

```
1 module pri_encoder (x, y);
2
3     input [3:0] x;
4     output reg [1:0] y;
5
6     always @(*) begin
7         casex (x)
8             4'b1xxx: y = 2'b11;
9             4'b01xx: y = 2'b10;
10            4'b001x: y = 2'b01;
11            4'b000x: y = 2'b00;
12        endcase
13    end
14
15 endmodule
```

3 N-Bit ALU

```
1 module Nbit_ALU (in0, in1, opcode, out);
2     parameter W = 4;
3
4     input [W-1:0] in0, in1;
5     input [1:0] opcode ;
6     output reg [W-1:0] out;
7
8     always @(*) begin
9         case (opcode)
10            2'b00: out = in0 + in1;
11            2'b01: out = in0 | in1;
12            2'b10: out = in0 - in1;
13            2'b11: out = in0 ^ in1;
14        endcase
15    end
16 endmodule
```

4 Decimal-to-BCD Encoder

```
1 module BCD_enc (D0, D1, D2, D3, D4, D5, D6, D7, D8, D9, Y0, Y1, Y2, Y3);
2
3 input D0, D1, D2, D3, D4, D5, D6, D7, D8, D9;
4
5
6 output reg Y0, Y1, Y2, Y3;
7
8 always @(*) begin
9     case ({D9,D8,D7,D6,D5,D4,D3,D2,D1,D0})
10
11         10'b0000000001 : {Y3,Y2,Y1,Y0} = 4'b0000;
12         10'b0000000010 : {Y3,Y2,Y1,Y0} = 4'b0001;
13         10'b00000000100 : {Y3,Y2,Y1,Y0} = 4'b0010;
14         10'b00000001000 : {Y3,Y2,Y1,Y0} = 4'b0011;
15         10'b00000010000 : {Y3,Y2,Y1,Y0} = 4'b0100;
16         10'b00000100000 : {Y3,Y2,Y1,Y0} = 4'b0101;
17         10'b00001000000 : {Y3,Y2,Y1,Y0} = 4'b0110;
18         10'b00010000000 : {Y3,Y2,Y1,Y0} = 4'b0111;
19         10'b00100000000 : {Y3,Y2,Y1,Y0} = 4'b1000;
20         10'b01000000000 : {Y3,Y2,Y1,Y0} = 4'b1001;
21
22         default : {Y3,Y2,Y1,Y0} = 4'b0000;
23
24     endcase
25 end
26
27 endmodule
```